मई May **17th Sunday**

Polar coords $\Rightarrow x = r\cos\theta$

$y = r\sin\theta$

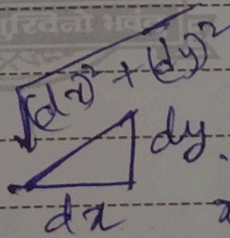
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

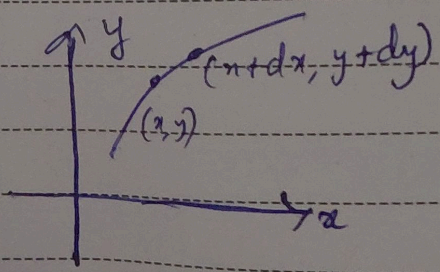
$$\Rightarrow \frac{\partial^2 v}{\partial r^2} + \frac{1}{r}\frac{\partial v}{\partial r} + \frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2} = 0$$

because $\frac{\partial}{\partial x} = \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x}$

मई May **18th Monday**

Calculus of Variance





$$\text{length} = \int_{x_1}^{x_2} \sqrt{(dx)^2 + (dy)^2}$$

$$\Rightarrow \int_{x_1}^{x_2} \sqrt{1 + (\dot{y})^2}\, dx$$

$$\Rightarrow \int_{x_1}^{x_2} f(y, \dot{y}, x)\, dx.$$

मई May **19**th Tuesday

Principle of variance

Optimal path $\Rightarrow$ $\boxed{\delta I = 0}$

where $I = \int_{x_1}^{x_2} \sqrt{1+(\dot{y})^2}\, dx$

$$\delta I = \delta \int_{x_1}^{x_2} \sqrt{(1+(\dot{y})^2)}\, dx = \delta \int_{x_1}^{x_2} f(y, \dot{y}, x)\, dx$$

$= 0$

मई May **20**th Wednesday

$$y(x, \alpha) = y(x, 0) + \alpha\eta(x)$$

$$I(\alpha) = \int_{x_1}^{x_2} f(y(x,\alpha), \dot{y}(x,\alpha), x)\, dx.$$

$\frac{\partial I}{\partial \alpha} = 0 \rightarrow$ extremum path

मई May 21st Thurs

Euler Lagra

$\frac{\partial I}{\partial \alpha} =$

y →

मई May 22

$\frac{\partial I}{\partial \alpha}$

मई May **21st Thursday**

Euler lagrange's eqⁿ

$$\frac{\partial I}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial \dot{y}}\frac{\partial \dot{y}}{\partial \alpha} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial \alpha} \right) dx$$

$$\left(\because\ df = \left(\frac{\partial f}{\partial x}\right)_{y,\dot{y}} dx + \left(\frac{\partial f}{\partial y}\right)_{x,\dot{y}} dy + \left(\frac{\partial f}{\partial \dot{y}}\right)_{x,y} d(\dot{y}) \right)$$

$y \to f^n$ of α, $x \to$ independent of α

मई May **22nd Friday**

$$\Rightarrow \boxed{\frac{\partial x}{\partial \alpha} = 0}$$

$$\frac{\partial I}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial \dot{y}}\frac{\partial \dot{y}}{\partial \alpha} \right) dx$$

$$= \int_{x_1}^{x_2} \frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha}\, dx + \underbrace{\int_{x_1}^{x_2} \frac{\partial f}{\partial \dot{y}}\frac{\partial^2 y}{\partial x\, \partial \alpha}\, dx}$$

Use fuv formula

मई May **23**rd Saturday

2nd Term

$$\Rightarrow \left.\frac{\partial f}{\partial \dot{y}} \frac{\partial y}{\partial \alpha}\right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial \dot{y}}\right) \frac{\partial y}{\partial \alpha}\, dx$$

0

$$\Rightarrow \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}}\right)\right] \frac{\partial y}{\partial \alpha}\, dx = 0$$

मई May **24**th Sunday

$$\Rightarrow \boxed{\frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}}\right)} \rightarrow \text{Euler Lagrange eq}^n$$

$$I = \int \sqrt{dx^2 + dy^2}$$

$$\delta I = 0 \Rightarrow \delta \int \sqrt{1 + (\dot{y})^2}\, dx = 0$$

$$f = \sqrt{1 + (\dot{y})^2}$$

मई May **25**th Monda

$$\frac{\partial f}{\partial y} = 0$$

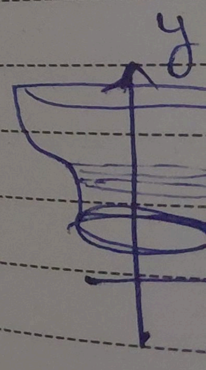
$$\frac{\partial f}{\partial \dot{y}} =$$

$$\frac{d}{dx}\left(\right.$$

$$\frac{\dot{y}}{\sqrt{1+\dot{y}^2}} =$$

मई May **26**th

Minimu



मई May **25th Monday**

$$\frac{\partial f}{\partial y} = 0$$

$$\frac{\partial f}{\partial \dot{y}} = \frac{\dot{y}}{\sqrt{1+\dot{y}^2}}$$

$$\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}}\right) = 0 \Rightarrow \frac{\partial f}{\partial \dot{y}} = c$$

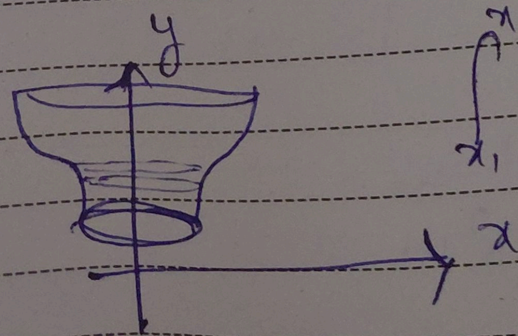
$$\frac{\dot{y}}{\sqrt{1+\dot{y}^2}} = c \Rightarrow \dot{y} = m$$

$$\Rightarrow y = mx + c'$$

मई May **26th Tuesday**

(straight line)

Minimum surface of revolution



$$\int_{x_1}^{x_2} 2\pi x \sqrt{dx^2 + dy^2}$$

$$\Rightarrow \int_{x_1}^{x_2} 2\pi x \sqrt{1+(\dot{y})^2}\, dx.$$

$$\Rightarrow f = 2\pi x \sqrt{1+(\dot{y})^2}$$

मई
May **27**th Wednesday

$$\Rightarrow \frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}}\right) \quad \left(\text{since } f = f(\dot{y}, x) \quad \frac{\partial f}{\partial y} = 0 \right)$$

$$\Rightarrow \cancel{\frac{\partial f}{\partial y} 0 = \frac{d}{dx}\Big(}$$

$$\Rightarrow \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}}\right) = 0$$

$$\Rightarrow \frac{\partial f}{\partial \dot{y}} = c$$

मई
May **28**th Thursday

$$\Rightarrow \frac{2\pi x \dot{y}}{\sqrt{1+(\dot{y})^2}} = C$$

$$\Rightarrow 4\pi^2 x^2 (\dot{y})^2 = c^2 + c^2(\dot{y})^2$$

$$\Rightarrow (\dot{y})^2 = \frac{c^2}{4\pi^2 x^2 - c^2}$$

$$= \frac{1}{\left(\frac{2\pi}{c}\right)^2 x^2 - 1}$$

$$\dot{y} = \sqrt{\frac{1}{mx^2 - 1}}$$

मई
May **29**th Fri

$\dot{y} =$

$dy =$

$y =$

मई
May **30**

$y =$

$y =$

मई May 29th Friday

$$\dot{y} = \sqrt{\frac{1}{mx^2 - 1}}$$

$$dy = \frac{dx}{\sqrt{mx^2-1}}$$

$$y = \int \frac{dx}{\sqrt{mx^2-1}} = c' \int \frac{dx}{\sqrt{x^2 - c^2}}$$

$$\cosh^2 x - \sinh^2 x = 1$$

मई May 30th Saturday

$$x = \cosh z.$$

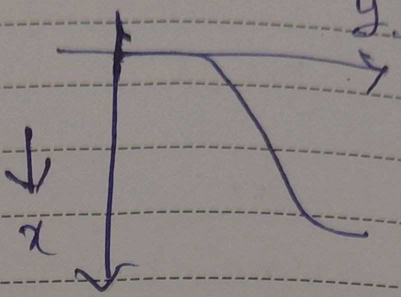
$$dx = \sinh z \, dz.$$

$$y = c' \int dz = c' z + a$$

$$y = \cosh^{-1}\left(\frac{x}{c'}\right) + a$$

मई
May **31**st Sunday

Brandistochrone problem



$$t = \int \frac{ds}{v}$$

$$= \int \sqrt{\frac{dx^2 + dy^2}{v^2}}$$

$$\Rightarrow v = \sqrt{2gx}$$

$$= \int \sqrt{\frac{dx^2 + dy^2}{2gx}}$$

जून
June **1**st Monday

$$= \int \sqrt{\frac{1 + (\dot{y})^2}{2gx}}\, dx$$

$$f = \sqrt{\frac{1 + (\dot{y})^2}{2gx}}$$

again $\frac{\partial f}{\partial y} = 0$

$\frac{\partial f}{\partial \dot{y}} = c$ again

जून June 2nd Tuesday

If $f = f(y_1, y_2, \dots, \dot{y}_1, \dot{y}_2, \dots, x)$

then $\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right) = 0$ for each y_i

Lagrangian eq^n

$$\Rightarrow \frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0.$$

जून June 3rd Wednesday

If a particular generalised co-ordinate q_k is absent in the lagrangian, then q_k is called cyclic coordinate.

$$\frac{\partial L}{\partial q_k} = 0 \Rightarrow \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = 0$$

$$\Rightarrow \frac{\partial L}{\partial \dot{q}_k} = \text{constant}$$

जून June **4th** Thursday

$q_k \rightarrow$ generalised coordinate

$\frac{\partial L}{\partial \dot{q}_k} \rightarrow$ generalised momentum

$\left\{ q_i , \frac{\partial L}{\partial \dot{q}_i} \right\} \rightarrow$ canonically conjugate variables

ex $\Rightarrow$ $\{x, p_x\}$, $\{t, E\}$, $\{\theta, p_\theta\}$

जून June **5th** Friday

$$T = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 \right) = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right)$$

$$\frac{\partial L}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}$$

जून June **6th** Saturday

Conservation

$L(q_i, \dot{q}$

$\frac{dL}{dt} \neq$

$\frac{dL}{dt} = \Sigma$

जून June **7th** Su

$=$

$\frac{d}{dt}$

जून June **6th Saturday**

Conservation of energy

$$\boxed{\frac{\partial L}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)}$$

$L(q_i, \dot{q}_i)$

$\frac{dL}{dt} \neq 0$, $\frac{\partial L}{\partial t} = 0$ when L does not explicitly depend on time

$$\frac{dL}{dt} = \sum_i \left[\frac{\partial L}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\right].$$

जून June **7th Sunday**

$$= \sum_i \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\dot{q}_i + \frac{\partial L}{\partial \dot{q}_i}\frac{d}{dt}(\dot{q}_i)\right]$$

$$= \sum_i \left[\frac{d}{dt}\left(\dot{q}_i\frac{\partial L}{\partial \dot{q}_i}\right)\right]$$

$$\frac{d}{dt}\left(L - \sum_i \dot{q}_i\frac{\partial L}{\partial \dot{q}_i}\right) = 0.$$

variable

जून June 8th Monday

$$\underbrace{L - \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i}}_{H \text{ (Hamiltonian)}} = \text{conserved}$$

if L is explicitly not dependent on time

$H \to$ conserved

जून June 9th Tuesday

$$H = \left(\sum_i p_i \dot{q}_i\right) - L \quad ; \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

$$\frac{\partial T}{\partial \dot{q}_i} = \frac{\partial L}{\partial \dot{q}_i} = p_i$$

$$T = \sum_{jk} a_{jk} \dot{q}_j \dot{q}_k$$

जून June 10th Wedn

Euler's theo

If f is

$\sum_j a_j$

$\sum_j \dot{q}_j$

जून June 11

H =

H

(poten

जून June 10th Wednesday

Euler's theorem

If f is a homogenous f^n of order n, then

$$\sum_j a_j \frac{\partial f}{\partial a_j} = nf.$$

$$\sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = 2T \qquad \left(T = \tfrac{1}{2} m v^2\right)$$

जून June 11th Thursday

$$H = \left(\sum p_i \dot{q}_i\right) - L = 2T - L \qquad (L = T - V)$$

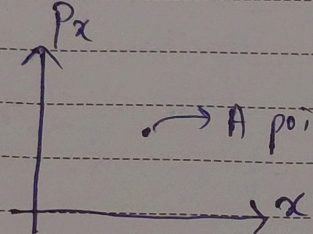
$$= 2T - (T - V)$$

$$= \underbrace{T + V}_{E} \rightarrow \text{conserved}$$

H represents total energy only if V (potential energy) doesn't depend on $\dot{q}_i$ (velocity)

जून June **12**th Friday

⇒ If we know pos^n & momentum of any classical body, we knows its state of motion.



→ A point in phase space represent state of motion of the object.

जून June **13**th Saturday

$$H(q_i, p_i) = \left(\sum_i p_i \dot{q}_i\right) - L(q_i, \dot{q}_i)$$

$$dH = \sum_i \frac{\partial H}{\partial q_i} dq_i + \sum_i \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt$$

$$dH = \sum_i p_i d\dot{q}_i + \sum_i \dot{q}_i dp_i - dL$$

$$\Rightarrow \sum \left(p_i d\dot{q}_i + \dot{q}_i dp_i\right) - dL$$

जून June **14**th Sunday

$\Rightarrow dH = \Sigma$

$$\frac{\partial H}{\partial q_i} =$$

जून June **15**th Mo

From varia

$$\delta \int_{t_1}^{t_2} L dt =$$

$$\Rightarrow \delta \int_{t_1}$$

$$\delta = d$$

14th Sunday
जून June

$$\Rightarrow dH = \sum_i (\cancel{p_i d\dot{q}_i} + \dot{q}_i dp_i)$$
$$- \left(\sum \dot{p}_i dq_i - \sum \cancel{p_i d\dot{q}_i} - \frac{\partial L}{\partial t} dt\right)$$

$$\frac{\partial H}{\partial q_i} = -\dot{p}_i \quad \Big| \quad \frac{\partial H}{\partial p_i} = \dot{q}_i$$

15th Monday
जून June

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

From variational principle,

$$\delta \int_{t_1}^{t_2} L\,dt = 0$$

$$\Rightarrow \delta \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}_i - H\right] dt = 0$$

$$\delta = d\alpha \cdot \frac{\partial}{\partial \alpha}$$

जून June

(a) $\Rightarrow$ $n(n-1)$

t_1 $\quad$ t_2

$\boxed{\sum l_i = n}$

$$(x + x^2 + \cdots + x^{k-1})^n \to x^n$$

जून June **19**th Friday

(Class continue)

$$\Rightarrow \left[d\alpha \int_{t_1}^{t_2} \left(p_i \frac{\partial \dot{q}_i}{\partial \alpha} + \dot{q}_i \frac{\partial p_i}{\partial \alpha} - \frac{\partial H}{\partial q_i} \frac{\partial q_i}{\partial \alpha} - \frac{\partial H}{\partial p_i} \frac{\partial p_i}{\partial \alpha} - \frac{\partial H}{\partial t} \frac{\partial t}{\partial \alpha} \right) \right]$$

$$= 0$$

We get $\frac{\partial H}{\partial q_i} = -\dot{p}_i$ & $\frac{\partial H}{\partial p_i} = \dot{q}_i$